

Problem 25.5

Annuity A offers to pay you \$100 per annum, convertible continuously, for the next five years. Annuity B offers you X at the end of each year for ten years. The annual effective interest rate i is 8%. Find X such that you are indifferent between the two annuities.

Solution

Since the annual effective interest rate is

$$i = 0.08,$$

the corresponding force of interest is

$$\delta = \ln(1 + i) = \ln(1.08) = 0.07696104.$$

Annuity A

Annuity A pays continuously at a rate of 100 per year for 5 years. Its present value is

$$100\bar{a}_{\overline{5}|}.$$

For a continuous annuity,

$$\bar{a}_{\overline{5}|} = \frac{1 - e^{-5\delta}}{\delta}.$$

Since $e^{-5\delta} = (1.08)^{-5}$,

$$PV_A = 100 \left(\frac{1 - (1.08)^{-5}}{\ln(1.08)} \right).$$

Thus,

$$PV_A = 415.0370087.$$

Annuity B

Annuity B pays X at the end of each year for 10 years. Its present value is

$$PV_B = Xa_{\overline{10}|0.08}.$$

Using

$$a_{\overline{10}|0.08} = \frac{1 - (1.08)^{-10}}{0.08},$$

we get

$$a_{\overline{10}|0.08} = 6.7100814.$$

For indifference, the present values must be equal:

$$PV_A = PV_B.$$

Therefore,

$$415.0370087 = X(6.7100814).$$

Solving for X ,

$$X = \frac{415.0370087}{6.7100814} = 61.852753.$$

Thus,

$$\boxed{X = 61.85}.$$